Interactive Science Notebook — Chapter 6 · Energy Transformations + Engineering Design

Trim each insert on the dashed line and glue into your notebook. Every interactive part has a flat, low-motor alternative.

INSERT 1 — Transfer vs Transformation · trim on the dashed line · glue the BACK

Transfer vs Transformation

Chapter 6 · Week 7

TWO WAYS ENERGY CHANGES THINGS

TRANSFER = energy MOVES from one object or place to another.

example: warm water warms a cool spoon (heat moves).

TRANSFORMATION = energy CHANGES FORM.

example: a flashlight changes stored energy into light.

BOTH = energy MOVES AND CHANGES FORM (most systems).

example: falling dominoes — motion moves AND some motion becomes sound.

MY OWN EXAMPLES

TRANSFER: _____

TRANSFORMATION: _____

BOTH: _____

INSERT 2 — Energy System Model · trim on the dashed line · glue the BACK

Energy System Model

Chapter 6 · Week 7

THE MODEL

INPUT
→ CHANGE IN FORM (TRANSFORMATION)
→ OUTPUT / OBSERVABLE CHANGE

DRAW ONE DEVICE

Pick one device (flashlight, buzzer, motor, wind-up toy). Draw it and label the chain.

[draw your device + label INPUT → TRANSFORMATION → OUTPUT]

MY CHAIN

INPUT: _____

TRANSFORMATION: _____

OUTPUT: _____

INSERT 3 — Device Detective · trim on the dashed line · glue the BACK

Device Detective

Chapter 6 · Week 7 · analyze one device

ANALYZE ONE DEVICE

1. What goes IN? _____

2. What CHANGES? _____

3. What comes OUT? _____

VERDICT (circle one)

TRANSFER TRANSFORMATION BOTH

WHY?

I circled this because _____

My evidence: I observed _____

INSERT 4 — Engineering Design · trim on the dashed line · glue the BACK

Engineering Design

Chapter 6 · Week 8

THE CYCLE

DESIGN → BUILD → TEST → REFINE

MY PROBLEM

Problem: _____

CRITERIA (what it must do)

1. _____

2. _____

3. _____

CONSTRAINTS (limits)

1. _____

2. _____

BLUEPRINT

Draw + label your device (trigger, output, energy change).

[draw + label your blueprint here]

INSERT 5 — Test → Change → Retest · trim on the dashed line · fold the flap DOWN and glue the BASE · flat option: write under the flap

Test → Change → Retest

Chapter 6 · Week 8 · lift the flap to check

TEST 1 (on the flap)

Did it trigger? ____ What output? ____ What failed? ____

WHAT WE CHANGED + WHY (under the flap)

We changed ONLY _____ because _____

TEST 2 RESULT

Did the change help? ____ Evidence: _____

MY ENERGY PATHWAY

INPUT → TRANSFORMATION → OUTPUT:

_____ → _____ → _____